

LAB Exercise – III Data Retrieval using JOINS, Subqueries and Correlated queries

Q1. A

```
mysql> CREATE TABLE Clients
-> (
->     client_id INT PRIMARY KEY,
->     client_name VARCHAR(50),
->     email VARCHAR(100),
->     city VARCHAR(50)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE Freelancers
-> (
->     freelancer_id INT PRIMARY KEY,
->     freelancer_name VARCHAR(50),
->     skill VARCHAR(50),
->     hourly_rate INT
-> );
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE TABLE Projects
-> (
->     project_id INT PRIMARY KEY,
->     client_id INT,
->     project_title VARCHAR(100),
->     budget INT,
->     status VARCHAR(30),
->     FOREIGN KEY(client_id) REFERENCES Clients(client_id)
-> );
Query OK, 0 rows affected (0.01 sec)

mysql> CREATE TABLE Bids
-> (
->     bid_id INT PRIMARY KEY,
->     project_id INT,
->     freelancer_id INT,
->     bid_amount INT,
->     bid_status VARCHAR(20),
->     FOREIGN KEY(project_id) REFERENCES Projects(project_id),
->     FOREIGN KEY(freelancer_id) REFERENCES Freelancers(freelancer_id)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE Payments
-> (
->     payment_id INT PRIMARY KEY,
->     project_id INT,
->     freelancer_id INT,
->     amount INT,
->     payment_date DATE,
->     FOREIGN KEY(project_id) REFERENCES Projects(project_id),
->     FOREIGN KEY(freelancer_id) REFERENCES Freelancers(freelancer_id)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO Clients
-> VALUES
-> (101,'Oshail','oshail@gmail.com','Delhi'),
-> (102,'Harshit','harshit@gmail.com','Bangalore'),
-> (103,'Kaira','kaira@gmail.com','Mumbai'),
-> (104,'Aadi','aadi@gmail.com','Pune'),
-> (105,'Parth','parth@gmail.com','Jaipur');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Freelancers
-> VALUES
-> (201,'Riya','Web Development',800),
-> (202,'Aman','Graphic Design',600),
-> (203,'Neha','Machine Learning',1200),
-> (204,'Rohan','Content Writing',500),
-> (205,'Arjun','Cyber Security',1500);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Projects
-> VALUES
-> (301,101,'E-Commerce Website',50000,'Completed'),
-> (302,102,'Company Logo Design',12000,'Completed'),
-> (303,103,'AI Chatbot',90000,'In Progress'),
-> (304,104,'SEO Content',15000,'Completed'),
-> (305,105,'Security Audit',70000,'Pending');
Query OK, 5 rows affected (0.00 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Bids
-> VALUES
-> (401,301,201,45000,'Accepted'),
-> (402,302,202,10000,'Accepted'),
-> (403,303,203,85000,'Pending'),
-> (404,304,204,14000,'Accepted'),
-> (405,305,205,68000,'Pending');
Query OK, 5 rows affected (0.00 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Payments
-> VALUES
-> (501,301,201,45000,'2026-01-10'),
-> (502,302,202,10000,'2026-01-15'),
-> (503,304,204,14000,'2026-01-20');
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Clients;
+-----+-----+-----+-----+
| client_id | client_name | email | city |
+-----+-----+-----+-----+
| 101 | Oshail | oshail@gmail.com | Delhi |
| 102 | Harshit | harshit@gmail.com | Bangalore |
| 103 | Kaira | kaira@gmail.com | Mumbai |
| 104 | Aadi | aadi@gmail.com | Pune |
| 105 | Parth | parth@gmail.com | Jaipur |
+-----+-----+-----+-----+
5 rows in set (0.01 sec)
```

```
mysql>
mysql> SELECT * FROM Freelancers;
+-----+-----+-----+-----+
| freelancer_id | freelancer_name | skill | hourly_rate |
+-----+-----+-----+-----+
| 201 | Riya | Web Development | 800 |
| 202 | Aman | Graphic Design | 600 |
| 203 | Neha | Machine Learning | 1200 |
| 204 | Rohan | Content Writing | 500 |
| 205 | Arjun | Cyber Security | 1500 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM Projects;
+-----+-----+-----+-----+-----+
| project_id | client_id | project_title | budget | status |
+-----+-----+-----+-----+-----+
| 301 | 101 | E-Commerce Website | 50000 | Completed |
| 302 | 102 | Company Logo Design | 12000 | Completed |
| 303 | 103 | AI Chatbot | 90000 | In Progress |
| 304 | 104 | SEO Content | 15000 | Completed |
| 305 | 105 | Security Audit | 70000 | Pending |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM Bids;
+-----+-----+-----+-----+-----+
| bid_id | project_id | freelancer_id | bid_amount | bid_status |
+-----+-----+-----+-----+-----+
| 401 | 301 | 201 | 45000 | Accepted |
| 402 | 302 | 202 | 10000 | Accepted |
| 403 | 303 | 203 | 85000 | Pending |
| 404 | 304 | 204 | 14000 | Accepted |
| 405 | 305 | 205 | 60000 | Pending |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM Payments;
+-----+-----+-----+-----+-----+
| payment_id | project_id | freelancer_id | amount | payment_date |
+-----+-----+-----+-----+-----+
| 501 | 301 | 201 | 45000 | 2024-01-10 |
| 502 | 302 | 202 | 10000 | 2024-01-15 |
| 503 | 304 | 204 | 14000 | 2024-01-20 |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

INNER JOIN

Display only projects that have clients.

SELECT

- > c.client_name,
- > p.project_title,
- > p.budget
- > FROM Clients c
- > INNER JOIN Projects p
- > ON c.client_id=p.client_id;

```
mysql> SELECT
-> c.client_name,
-> p.project_title,
-> p.budget
-> FROM Clients c
-> INNER JOIN Projects p
-> ON c.client_id=p.client_id;
+-----+-----+-----+
| client_name | project_title | budget |
+-----+-----+-----+
| Oshail | E-Commerce Website | 50000 |
| Harshit | Company Logo Design | 12000 |
| Kaira | AI Chatbot | 90000 |
| Aadi | SEO Content | 15000 |
| Parth | Security Audit | 70000 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

LEFT JOIN

Justification: Shows every freelancer even if they have not received any payment.

SELECT

- > f.freelancer_name,
- > pay.amount
- > FROM Freelancers f
- > LEFT JOIN Payments pay
- > ON f.freelancer_id=pay.freelancer_id;

```
mysql> SELECT
-> f.freelancer_name,
-> pay.amount
-> FROM Freelancers f
-> LEFT JOIN Payments pay
[ -> ON f.freelancer_id=pay.freelancer_id;
+-----+-----+
| freelancer_name | amount |
+-----+-----+
| Riya            | 45000  |
| Aman            | 10000  |
| Neha            | NULL   |
| Rohan           | 14000  |
| Arjun           | NULL   |
+-----+-----+
5 rows in set (0.00 sec)
```

RIGHT JOIN

Justification: Shows every payment record even if freelancer details are unavailable.

SELECT

```
-> f.freelancer_name,
-> pay.amount
-> FROM Freelancers f
-> RIGHT JOIN Payments pay
-> ON f.freelancer_id=pay.freelancer_id;
```

```
mysql> SELECT
-> f.freelancer_name,
-> pay.amount
-> FROM Freelancers f
-> RIGHT JOIN Payments pay
[ -> ON f.freelancer_id=pay.freelancer_id;
+-----+-----+
| freelancer_name | amount |
+-----+-----+
| Riya            | 45000  |
| Aman            | 10000  |
| Rohan           | 14000  |
+-----+-----+
3 rows in set (0.00 sec)
```

FULL OUTER JOIN

Justification: Displays all freelancers and all payment records.

SELECT

```
-> f.freelancer_name,
-> pay.amount
-> FROM Freelancers f
-> LEFT JOIN Payments pay
-> ON f.freelancer_id=pay.freelancer_id
-> UNION
-> SELECT
-> f.freelancer_name,
-> pay.amount
-> FROM Freelancers f
-> RIGHT JOIN Payments pay
```

-> ON f.freelancer_id=pay.freelancer_id;

```
mysql> SELECT
-> f.freelancer_name,
-> pay.amount
-> FROM Freelancers f
-> LEFT JOIN Payments pay
-> ON f.freelancer_id=pay.freelancer_id
->
-> UNION
->
-> SELECT
-> f.freelancer_name,
-> pay.amount
-> FROM Freelancers f
-> RIGHT JOIN Payments pay
-> ON f.freelancer_id=pay.freelancer_id;
```

freelancer_name	amount
Riya	45000
Aman	10000
Neha	NULL
Rohan	14000
Arjun	NULL

5 rows in set (0.01 sec)

B.

INNER JOIN

Displays only the projects that have a matching client.

LEFT JOIN

Displays all freelancers, even if they have not received any payment.

RIGHT JOIN

Displays all payment records, even if there is no matching freelancer.

FULL OUTER JOIN

Displays all freelancers and all payment records, whether they have matching records or not.

Q2

1. Find freelancers earning more than average.

```
SELECT freelancer_name, hourly_rate
-> FROM Freelancers
-> WHERE hourly_rate >
-> (
-> SELECT AVG(hourly_rate)
-> FROM Freelancers
-> );
```

2. Find projects having budget greater than average.

```
SELECT project_title, budget
-> FROM Projects
-> WHERE budget >
-> (
-> SELECT AVG(budget)
-> FROM Projects
-> );
```

3. Find clients who have posted projects.

```
SELECT client_name
-> FROM Clients
-> WHERE client_id IN
-> (
-> SELECT client_id
-> FROM Projects
-> );
```

4. Find freelancers whose bid is accepted.

```
SELECT freelancer_name
-> FROM Freelancers
-> WHERE freelancer_id IN
-> (
-> SELECT freelancer_id
-> FROM Bids
-> WHERE bid_status='Accepted'
-> );
```

5. Find projects that have received payments.

```
SELECT project_title
-> FROM Projects
-> WHERE project_id IN
-> (
-> SELECT project_id
-> FROM Payments
-> );
```



```
mysql> SELECT freelancer_name, hourly_rate
-> FROM Freelancers
-> WHERE hourly_rate >
-> (
-> SELECT AVG(hourly_rate)
-> FROM Freelancers
[ -> );
```

freelancer_name	hourly_rate
Neha	1200
Arjun	1500

2 rows in set (0.05 sec)

```
mysql> SELECT project_title, budget
-> FROM Projects
-> WHERE budget >
-> (
-> SELECT AVG(budget)
-> FROM Projects
-> );
```

project_title	budget
E-Commerce Website	50000
AI Chatbot	90000
Security Audit	70000

3 rows in set (0.01 sec)

```
mysql> SELECT client_name
-> FROM Clients
-> WHERE client_id IN
-> (
-> SELECT client_id
-> FROM Projects
[ -> );
```

client_name
Oshail
Harshit
Kaira
Aadi
Parth

5 rows in set (0.00 sec)

```
mysql> SELECT freelancer_name
      -> FROM Freelancers
      -> WHERE freelancer_id IN
      -> (
      -> SELECT freelancer_id
      -> FROM Bids
      -> WHERE bid_status='Accepted'
[      -> );
```

```
+-----+
| freelancer_name |
+-----+
| Riya           |
| Aman           |
| Rohan          |
+-----+
```

3 rows in set (0.00 sec)

```
mysql> SELECT project_title
      -> FROM Projects
      -> WHERE project_id IN
      -> (
      -> SELECT project_id
      -> FROM Payments
[      -> );
```

```
+-----+
| project_title   |
+-----+
| E-Commerce Website |
| Company Logo Design |
| SEO Content     |
+-----+
```

3 rows in set (0.01 sec)

B. Compound Query

Displays complete information about completed freelance projects including client, freelancer, bid amount, and payment.

SELECT

```
-> c.client_name,
-> p.project_title,
-> f.freelancer_name,
-> b.bid_amount,
-> pay.amount,
-> pay.payment_date
-> FROM Clients c
-> INNER JOIN Projects p
-> ON c.client_id=p.client_id
-> INNER JOIN Bids b
-> ON p.project_id=b.project_id
-> INNER JOIN Freelancers f
-> ON b.freelancer_id=f.freelancer_id
-> INNER JOIN Payments pay
-> ON pay.project_id=p.project_id
-> AND pay.freelancer_id=f.freelancer_id
-> WHERE p.status='Completed';
```



```
mysql> SELECT
-> c.client_name,
-> p.project_title,
-> f.freelancer_name,
-> b.bid_amount,
-> pay.amount,
-> pay.payment_date
-> FROM Clients c
-> INNER JOIN Projects p
-> ON c.client_id=p.client_id
-> INNER JOIN Bids b
-> ON p.project_id=b.project_id
-> INNER JOIN Freelancers f
-> ON b.freelancer_id=f.freelancer_id
-> INNER JOIN Payments pay
-> ON pay.project_id=p.project_id
-> AND pay.freelancer_id=f.freelancer_id
-> WHERE p.status='Completed';
```

client_name	project_title	freelancer_name	bid_amount	amount	payment_date
Oshail	E-Commerce Website	Riya	45000	45000	2026-01-10
Harshit	Company Logo Design	Aman	10000	10000	2026-01-15
Aadi	SEO Content	Rohan	14000	14000	2026-01-20

3 rows in set (0.00 sec)